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AGI JOBS × OPENAI GYM × SUCCESSOR Ω

Evidence and Reproducibility Volume

*Synthetic benchmark constitutions, complete succession cycles,
sensitivity analysis, failure evidence and exact rebuild protocols*

NO SPECIALIST ASI BY NARRATIVE

This volume discloses the synthetic evidence programme and reproducibility contract. It tests whether the institution can route algorithms, compare complete architectures, reject hard-gate failures, choose simpler systems when justified, preserve negative findings and reproduce every disclosed result.

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This companion is part of the same versioned publication architecture. The flagship paper presents the central argument and essential results. This volume preserves the deeper specification or evidence needed for implementation, reproduction and adversarial review. All state claims remain bounded by the same rule : authority never exceeds fresh proof.

Canonical progression

Successor Manifold → Mission Advantage Gradient → SEIZE Underwriting → Bounded AGI Jobs → Formation Gym → Exact Release Freeze → Fresh Proof → Specialist ASI → Successor Ω .

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PART I

Evidence and Reproducibility

Volume

*Synthetic evidence proves execution and selection logic;
protected reality remains the gate for Specialist ASI and
authority*

Synthetic benchmarks, complete succession cycles, sensitivity analysis, environment cards
and exact rebuild requirements.

SUCCESSOR MANIFOLD → GRADIENT → SEIZE → BOUNDED JOBS → SPECIALIST ASI → SUCCESSOR Ω

1 Synthetic Evidence : Algorithm Sophistication Is Not Alpha

1.1 Evidence status

This section reports disclosed, deterministic synthetic results from the GoalOS v8.0.0-UVSI3 Sovereign Alpha Foundry reference implementation. The results demonstrate executable mechanics : mission compilation, routing, training, fresh-seed separation, hard gates, conservative Alpha accounting and refusal. They are not claims of real customer value, frontier-scale MuZero performance or production Specialist ASI.

The dependency-free core compares seven candidate classes :

1. Incumbent Workflow;
2. Deterministic Rules;
3. Behaviour Cloning;
4. Tabular Q-Learning;
5. Dyna-Q;
6. MuZero-Lite;
7. Shielded MuZero-Lite.

The MuZero-Lite candidates are transparent tabular representation/dynamics/prediction/search references inspired by MuZero, not neural reproductions at research-lab scale.

1.2 Three mission environments

Mission	Structural character	Why it matters
Invoice Integrity	Formal rules, recurring evidence patterns, reversible holds	Tests whether a simple transparent architecture dominates advanced planning.
Portfolio Building Operations	Partial observability, sequential diagnostics, delayed physical consequences, safety constraints	Strongest case for world models, planning and external action shields.
Grid Storage Resilience	Sequential operating choices, hidden conditions, delayed rewards, service continuity	Tests whether historical imitation or planning creates the best conservative result.

1.3 Results

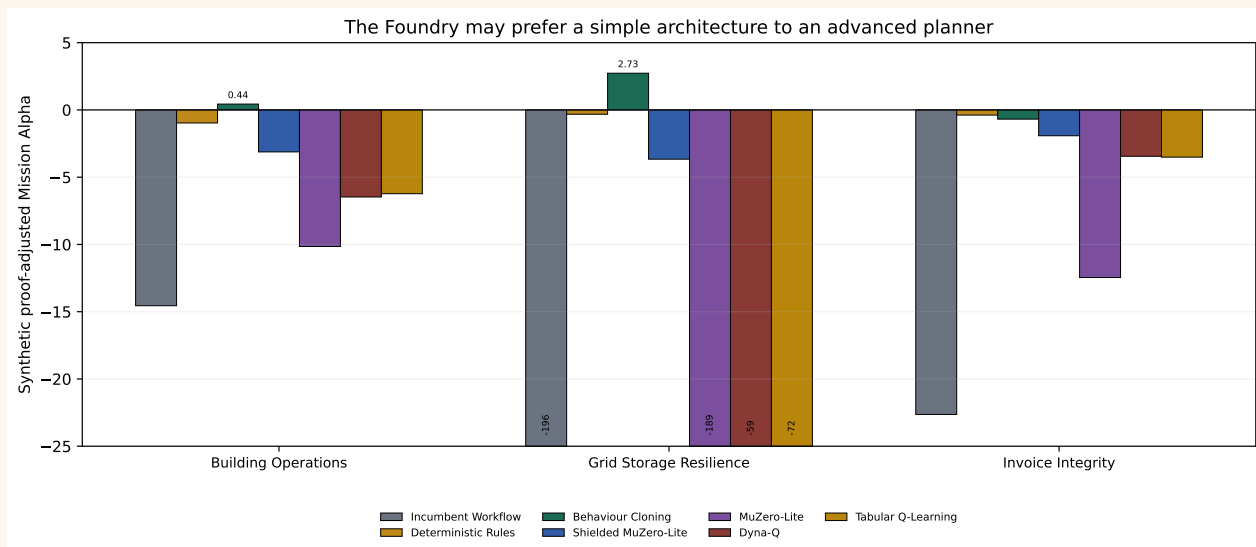


FIGURE 1. Synthetic proof-adjusted Mission Alpha by mission and candidate. Large negative values are clipped visually and annotated; the underlying records preserve exact values.

The disclosed reference outcomes are intentionally non-monotone :

- **Invoice Integrity** : no learning candidate produces positive proof-adjusted Alpha after hard gates and basis-risk reserve; the Foundry retains or repairs the Deterministic Rules reference.
- **Building Operations** : Behaviour Cloning produces the highest positive conservative Alpha in the disclosed run, while MuZero-Lite fails to justify its burden.
- **Grid Storage Resilience** : Behaviour Cloning again wins in the disclosed run; the more complex model-based candidates do not dominate after transfer and proof burden.

These results are not evidence that behaviour cloning is generally superior. They establish a more important institutional property : the Foundry can reject an advanced algorithm when the mission, data or learned model does not support its burden.

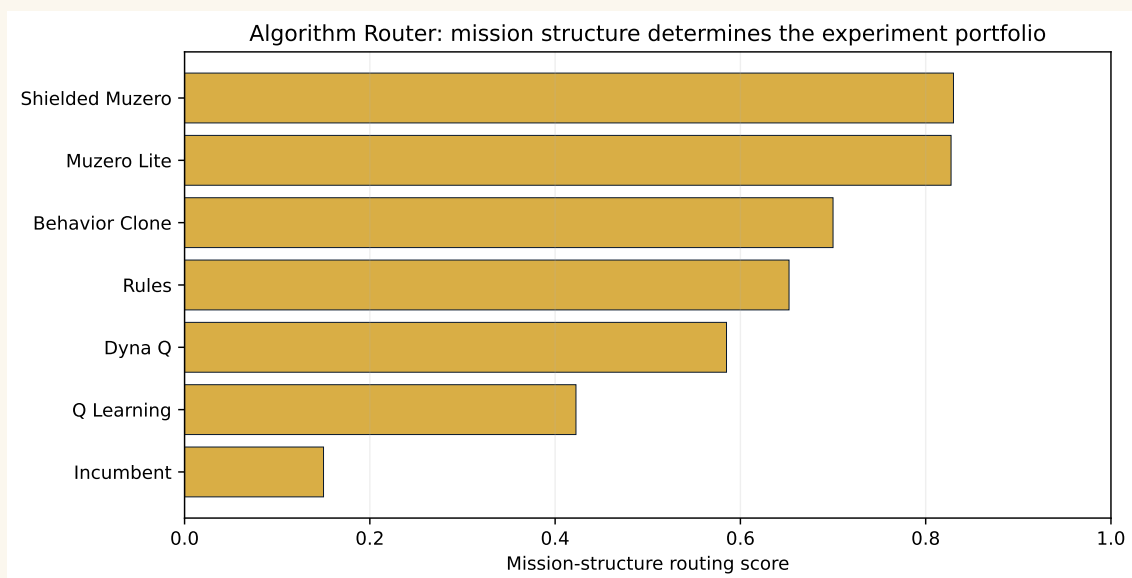


FIGURE 2. The Building Operations Router gives MuZero-style candidates strong formation priority because hidden dynamics and delayed consequences matter. Routing priority is not a proof verdict.

1.4 Hard gates

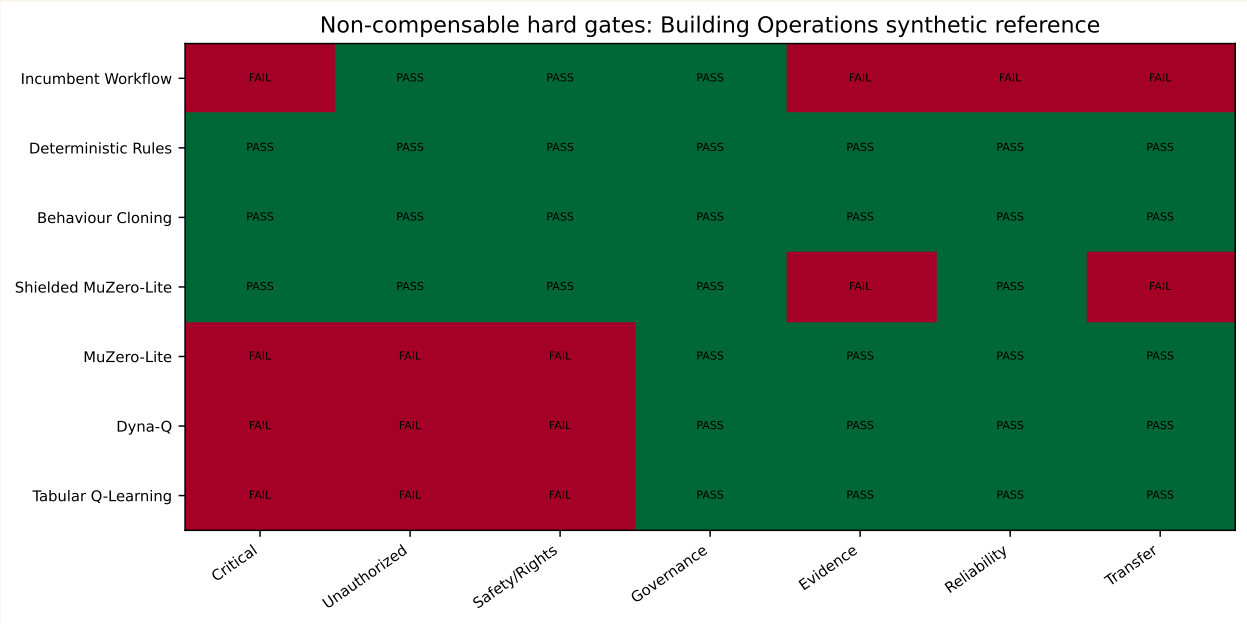


FIGURE 3. Synthetic non-compensable gates. Average reward cannot rescue a candidate that misses critical cases, acts without authority or fails evidence, reliability or transfer thresholds.

1.5 Institutional inference

The synthetic evidence supports four narrow conclusions :

- 1. algorithm routing and algorithm selection are different decisions ;
- 2. the mission may rationally justify testing MuZero while ultimately selecting something simpler ;
- 3. hard gates and basis-risk reserve can reverse the development ranking ;
- 4. a training result can create candidacy but not authority.

Synthetic evidence conclusion

A Foundry that always selects MuZero is not an Alpha Foundry. It is an algorithm marketing system. The institutional test is whether the complete candidate creates the strongest conservative mission advantage after proof, cost, risk, transfer and governance are counted.

2 AJGIB-1 : A Synthetic AGI Job Graph Integrity Benchmark

2.1 Purpose and claim boundary

AJGIB-1 evaluates whether typed AGI Job Graphs improve institutional completeness, evidence coverage, authority control and conservative Mission Alpha relative to monolithic or weakly structured alternatives. The benchmark is synthetic, deterministic, unaudited and intentionally simplified. It does not establish a real Specialist ASI, real customer savings or production authority. Its purpose is to make the paper’s contract, graph and proof logic executable and falsifiable.

The benchmark fixes seed 20260812 and evaluates five architecture classes across three mission environments, with 1,200 episodes per mission–architecture pair.

2.2 Mission environments

Mission	Synthetic structure
---------	---------------------

Invoice Integrity	High observability, dense deterministic rules, moderate criticality and a strong premium on provenance, duplicate detection and bounded payment holds.
Building Operations	Partial observability, delayed consequences, sensor uncertainty, physical tail risk and sequential diagnostic or intervention choices.
Acquisition Underwriting	Weak counterfactual ground truth, heterogeneous evidence, long causal chains, adversarial thesis formation and high human-review burden.

2.3 Architecture classes

The benchmark compares :

1. **Monolithic Agent** : one broadly capable policy with minimal typed decomposition;
2. **Flat Task List** : several named tasks without typed evidence dependencies or lifecycle closure;
3. **Typed Job Graph** : sealed contracts, explicit evidence paths, bounded capabilities and rollback;
4. **Typed Graph + Verifier** : the typed graph plus independent verification and challenge jobs;
5. **Adaptive Contract Foundry** : a quality-diversity portfolio that may modify job decomposition and architecture during formation, then freezes one exact challenger.

Each complete candidate is scored on fresh mission value, evidence coverage, reliability, governance, transfer, total institutional burden, Gym basis risk, critical failures and attempted authority leakage. The benchmark applies gate-normalized zero-tolerance logic to critical authority and reliability conditions while retaining all raw episode records.

2.4 Proof-adjusted architecture frontier

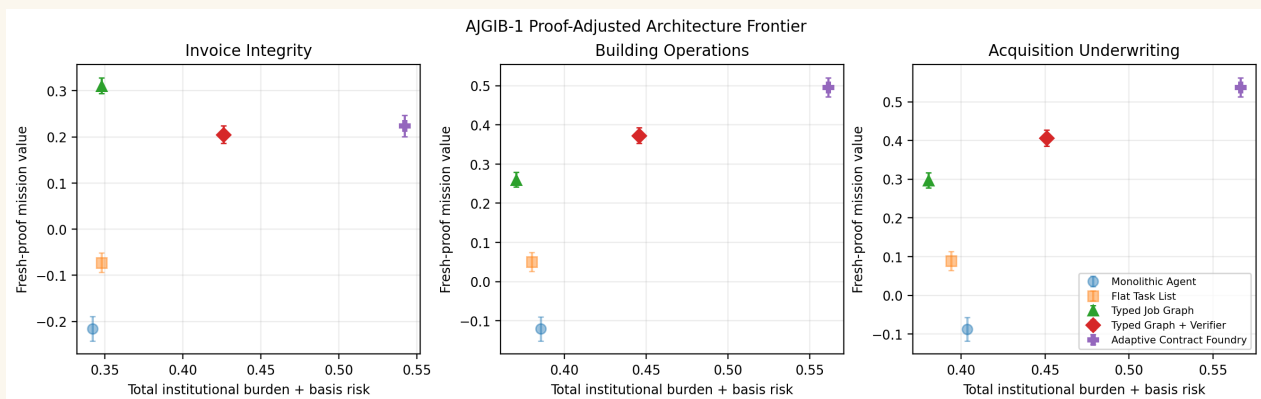


FIGURE 4. AJGIB-1 proof-adjusted architecture frontier. The preferred region combines higher fresh-proof mission value with lower total institutional burden and basis risk. Error bars are synthetic sampling intervals, not real-world confidence claims.

The frontier illustrates three principles. First, monolithic and flat architectures may appear inexpensive while losing value through weak evidence and governance. Second, the most complex candidate does not necessarily dominate a structured simpler candidate. Third, the location of the efficient frontier changes with mission structure.

2.5 Conservative Alpha after hard gates

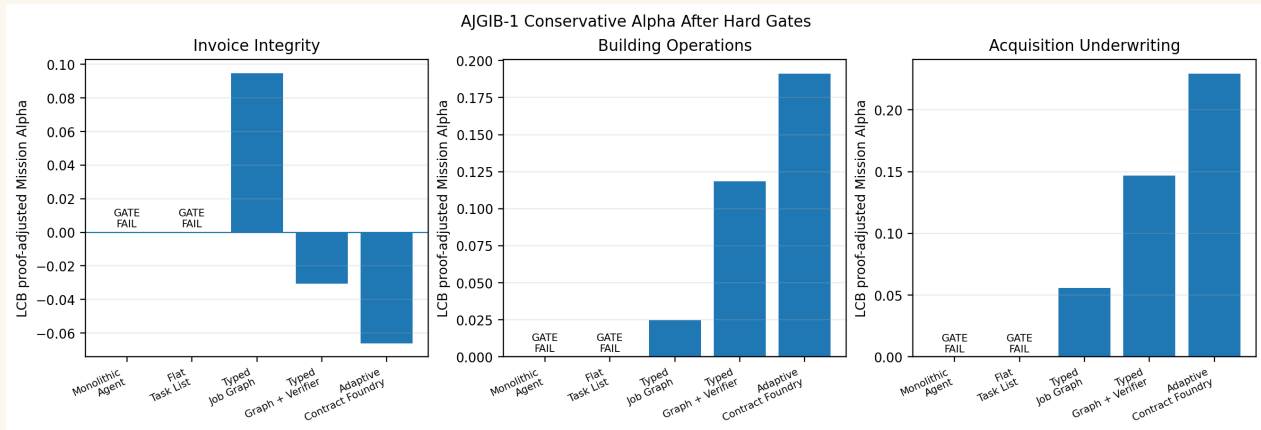


FIGURE 5. Lower-confidence-bound proof-adjusted Mission Alpha after gate normalization. A zero bar marked Gate Fail is inadmissible regardless of development performance.

In the disclosed synthetic run, the **Typed Job Graph** wins Invoice Integrity because its simpler structure captures the available value without the additional burden of a more adaptive Foundry. The **Adaptive Contract Foundry** wins Building Operations and Acquisition Underwriting because those missions benefit more from richer architecture search and sequential adaptation. This result is deliberately mission-dependent : the benchmark is designed to reject the thesis that more complexity is always better.

2.6 Complete institution score matrix

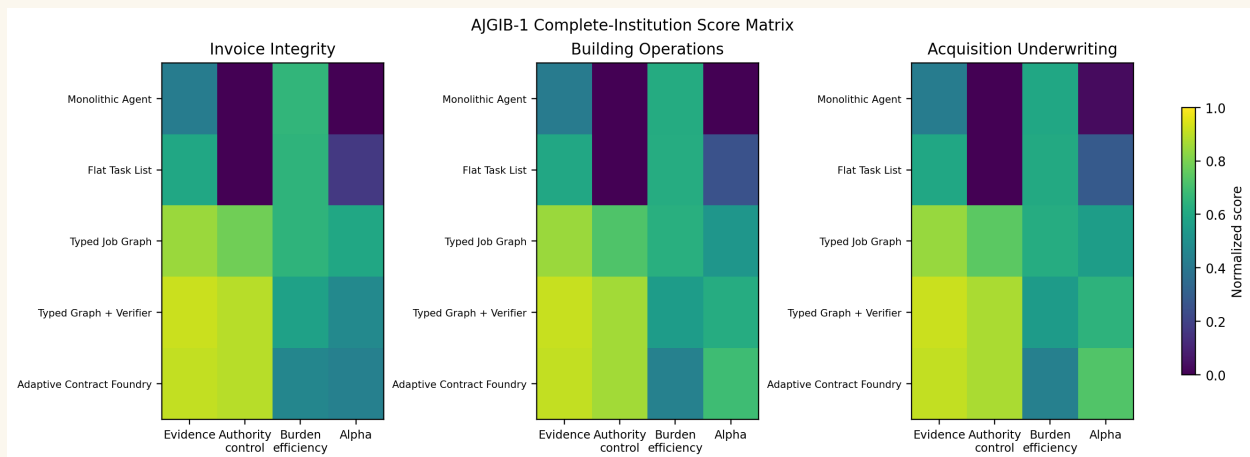


FIGURE 6. Normalized complete-institution dimensions. Evidence, authority control, burden efficiency and conservative Alpha are displayed separately rather than hidden inside one reward.

2.7 Reproducible outputs

The Evidence Companion includes :

- `rebuild.py`, the dependency-light deterministic generator;
- `manifest.json`, containing seed, mission parameters, architecture coordinates and claim boundary;
- `summary.csv` and `summary.json`, containing all mission–architecture aggregates;
- `episodes_sample.csv`, containing disclosed case-level records;
- `winners.json`, containing the gate-normalized mission winners;
- the three exact figures reproduced above.

2.8 What AJGIB-1 establishes

AJGIB-1 establishes only that the formal rules in this paper can be operationalized in a deterministic reference environment. It demonstrates that :

- typed job dependencies can be scored as institutional architecture rather than task prose;
- authority leakage and critical omissions can be treated as non-compensable gates;
- complete burden and Gym basis risk can reverse a naive development ranking;
- a simpler typed graph can defeat a more adaptive architecture on a sufficiently structured mission;
- the winning architecture can change when observability, sequential depth and criticality change.

It does not establish that the synthetic parameterization accurately represents any external company, market, building, acquisition or AI system. A real proof requires rights-cleared mission data, an exact incumbent, a protected case custodian, independently reviewed scorers and a bounded canary.

Synthetic evidence law

A synthetic benchmark may prove that the institution can execute its own rules. It cannot prove that a candidate dominates reality. Synthetic evidence creates candidacy for empirical proof, not inherited authority.

3 SJX-5 : Five-Mission Synthetic Triadic Benchmark

3.1 Purpose and claim boundary

SJX-5 tests whether the Successor $\Omega \times$ AGI Jobs doctrine can select different complete architectures under different mission structures. It is synthetic, deterministic, disclosed and unaudited. It does not establish real-world Specialist ASI, customer value or production authority.

The benchmark contains five mission families, seven candidate architectures, 800 protected synthetic cases per mission-candidate pair and 28,000 total case records. Each candidate is evaluated on fresh mission value, evidence, reliability, transfer, sovereignty, complete cost, human burden, basis risk, critical errors and unauthorized actions.

3.2 Mission and candidate set

Mission	Decision structure
Invoice Integrity	Strong rules, exact documents, duplicate and vendor controls; tests whether transparent deterministic architecture is sufficient.
Building Operations	Partial observability, sequential diagnostics and delayed physical consequences; tests world-model planning.
Grid Resilience	Hidden operating conditions, continuity constraints and tail risk; tests sovereign hybridization and bounded control.
Verified Software	Executable outputs, tests, repair, review and rollback; tests human-machine and hybrid institutions.
Acquisition Underwriting	Evidence acquisition, thesis and anti-thesis, delayed outcomes and high human authority; tests hybrid institutional intelligence.

The candidate set is : Incumbent, Deterministic Rules, Human-Machine, Behaviour Cloning, Model-Free RL, MuZero/-World Model and Sovereign Hybrid.

3.3 Promotion rule

For each candidate x , SJX-5 estimates a 95% lower confidence bound on fresh-work mission value and subtracts complete normalized burden and Gym-to-reality reserve :

$$\hat{\alpha}_m^{\text{SJX}}(x) = \text{LCB}_{0.95}[\hat{V}_m(x)] - V_m^{\text{ref}} - B_m(x) - \hat{\varepsilon}_{\text{gym}}(x). \quad (1)$$

A candidate is admissible only when it has zero critical errors, zero unauthorized actions and passes evidence, reliability, transfer and sovereignty thresholds. The highest development score does not win when a hard gate fails.

3.4 Results

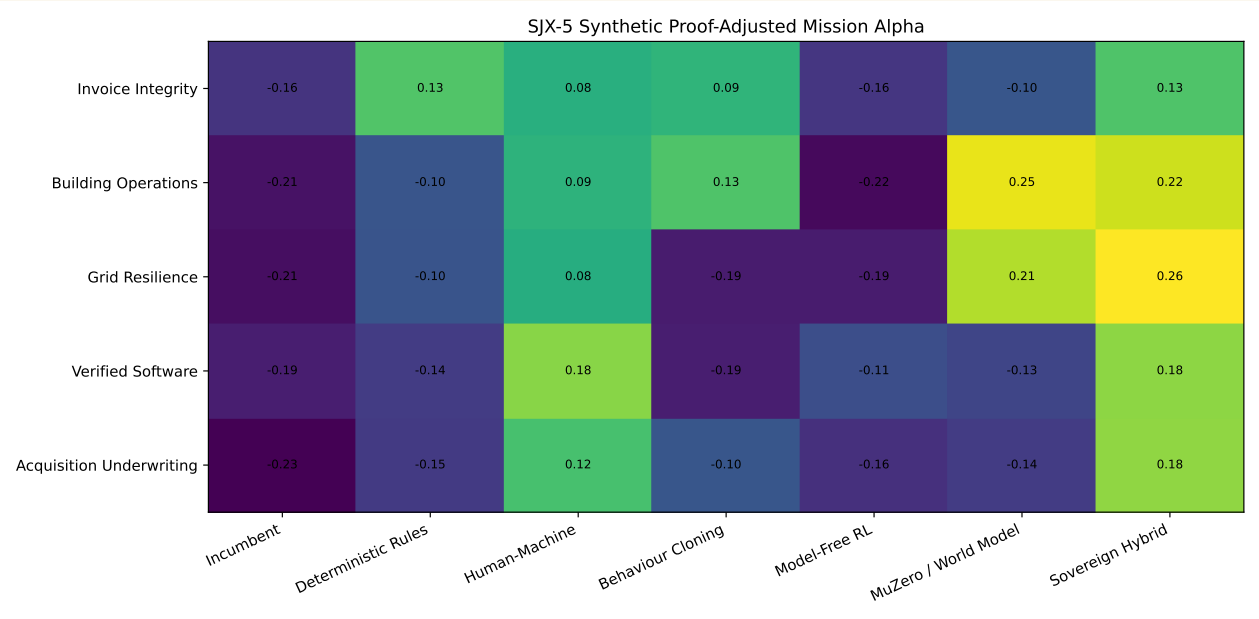


FIGURE 7. SJX-5 synthetic proof-adjusted Mission Alpha. The champion differs by mission; algorithm prestige is not a selection rule.

Mission	Synthetic champion	Proof-adjusted Alpha
Invoice Integrity	Deterministic Rules	0.129
Building Operations	MuZero / World Model	0.246
Grid Resilience	Sovereign Hybrid	0.263
Verified Software	Sovereign Hybrid	0.180
Acquisition Underwriting	Sovereign Hybrid	0.182

Invoice Integrity selects a simpler deterministic architecture because its transparent rules, evidence coverage and low burden outweigh the additional sophistication of learned systems. Building Operations selects MuZero/World Model because sequential planning across hidden dynamics creates enough value to survive cost and basis-risk deductions. Grid Resilience, Verified Software and Acquisition Underwriting select Sovereign Hybrid architectures because the complete institution—not one algorithm—creates the strongest conservative result.

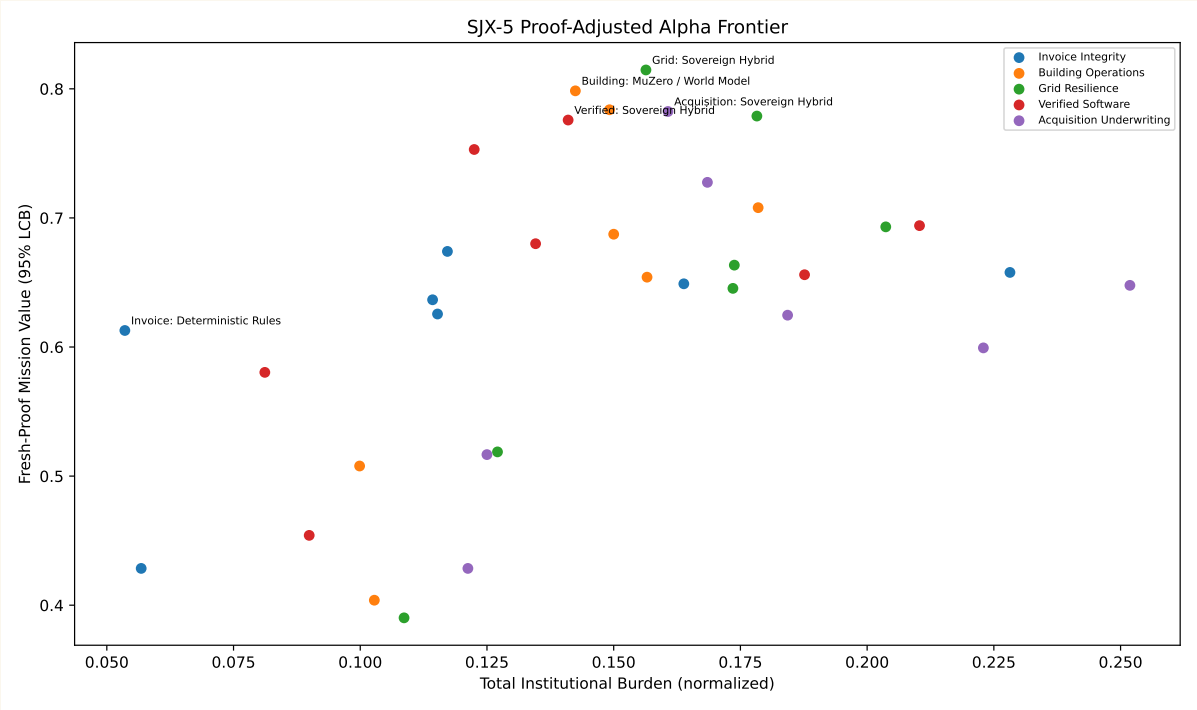


FIGURE 8. SJX-5 institutional frontier: Fresh-proof mission value is evaluated jointly with total institutional burden.

3.5 Hard gates

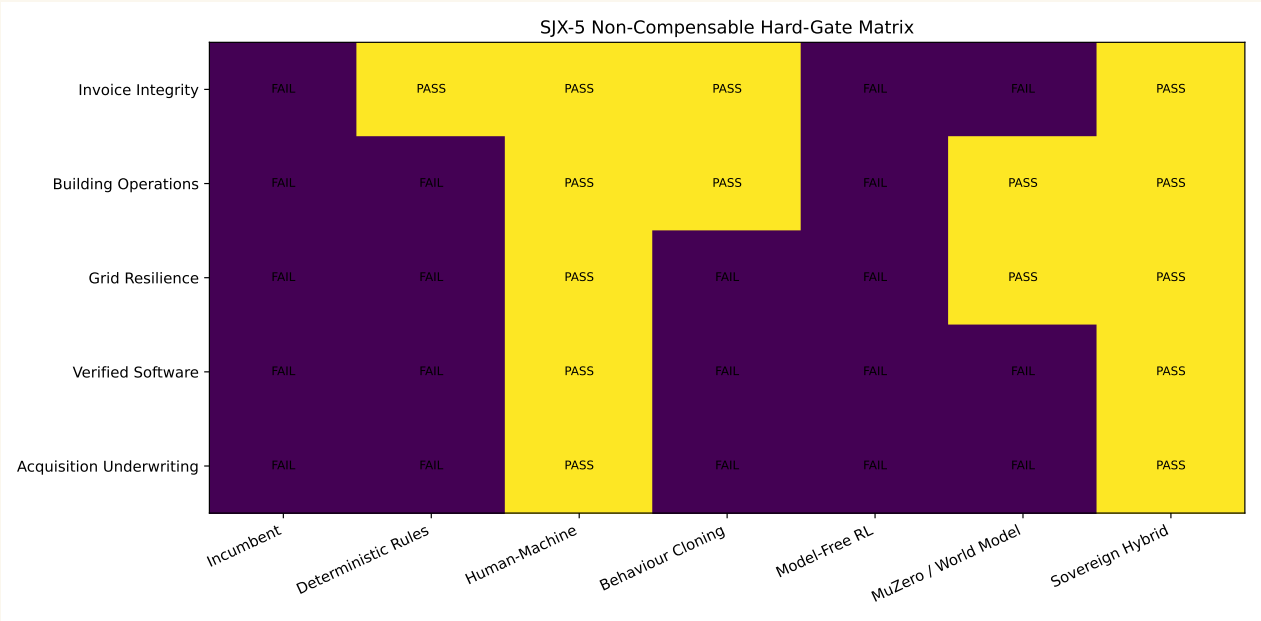


FIGURE 9. Non-compensable hard gates. Several high-scoring candidates are rejected because one critical error, unauthorized action or evidence failure cannot be traded against average utility.

3.6 What the benchmark establishes

The disclosed synthetic evidence supports five narrow conclusions :

1. the same algorithm should not be presumed superior across missions;
2. a simple architecture can rationally defeat a frontier-style planner;
3. MuZero-style planning can rationally win when hidden dynamics and sequential decisions justify its burden;
4. the complete institution can outperform every isolated algorithm through verifier, workflow, data, rights, rollback

- and human authority;
5. training results create candidacy only; no synthetic verdict creates real Specialist ASI or authority.

SJX 5 conclusion

A sovereign Foundry must be able to prove MuZero wins, prove something simpler wins, preserve a reserve, repair the incumbent or stop. Any system that always promotes the most advanced algorithm is optimizing narrative prestige rather than conservative Mission Alpha.

4 Synthetic Evidence Programme and Sensitivity Analysis

4.1 Why synthetic evidence is useful—and insufficient

Synthetic data can exercise the complete succession machinery before a rights-cleared customer mission is available. It can test partitioning, hidden ground truth, candidate formation, protected proof, authority masking, proof receipts, canary operation, impairment, rollback and successor-to-successor requalification. It cannot establish that the same utility function, data-generating process, human behaviour, operational cost or tail risk holds in production.

The v3.0 evidence programme therefore treats synthetic evidence as a signed *mechanism demonstration*. It supports claims about reproducibility and internal consistency of the reference cycle, not customer value, universal capability or production authority.

The results in this section are generated from the Aurelia reference environment. They are unaudited, non-customer, non-investment and non-production results. Their purpose is to expose the proof chain and its sensitivity, not to simulate away the need for real protected work.

4.2 Dataset partition and cryptographic commitments

The reference environment contains 4,300 generated cases. Formation receives visible labels ; protected proof, canary and transfer labels are separately held. The release includes SHA-256 commitments to the protected files before the published outcome record.

TABLE 2. Aurelia synthetic evidence partitions.

Partition	Cases	Candidate visibility	Institutional purpose
Formation	2,200	Ground truth visible to formation tooling	Architecture search, debugging, quality diversity and failure repair.
Protected proof	1,200	Inputs visible; labels and scorer protected	Matched challenger-versus-incumbent mission-dominance test.
Canary	400	Bounded operational inputs; hidden outcome record	Reversible A3 demonstration and complete-denominator value measurement.
Transfer	500	New anomaly mix and higher novelty	Chronicle transfer, provider-change impairment and next-successor proof.

The protected commitments are part of the Research Companion and Mission Pack. A production institution should commit not only the case file but the scorer version, threshold configuration, candidate manifest, environment image and permissible human-intervention policy.

4.3 Formation frontier on the Successor Manifold

The candidate set illustrates why the complete successor must be compared rather than the model alone. The deterministic repair is inexpensive and highly auditable but weak on novel bundled anomalies. The general AI copilot is fast but has low evidence custody and verifier independence. The specialist cell improves domain reasoning but retains substantial human burden. The sovereign hybrid combines evidence graph, deterministic controls, specialist

intelligence, critic, verifier, governor, Chronicle and portability.

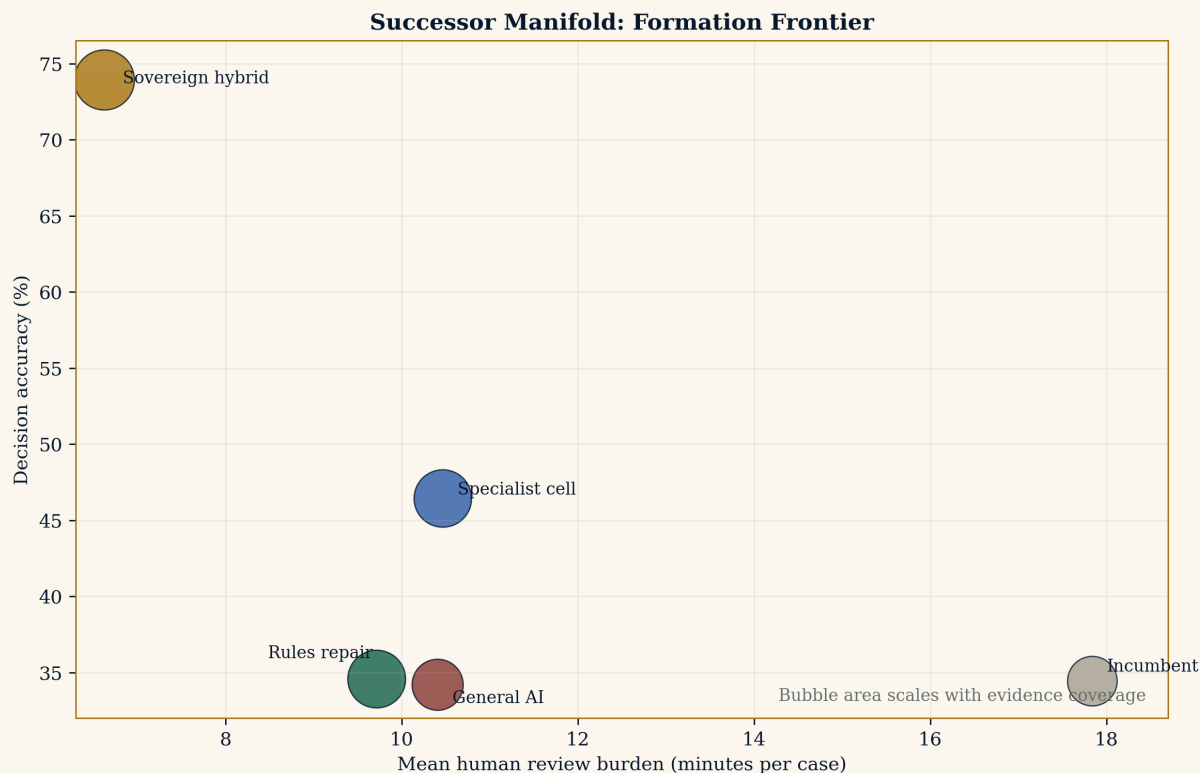


FIGURE 10. Synthetic formation frontier. Horizontal position is mean human-review burden ; vertical position is decision accuracy ; bubble area scales with evidence coverage. The plot is descriptive of the reference environment, not a universal ranking of architecture classes.

The frontier demonstrates a central GoalOS principle : the winning point is not merely the most accurate policy. It is the complete architecture whose mission gain survives evidence, governance, cost, human burden and authority constraints.

4.4 Recursive Foundry evidence

Seven bounded generations increase champion formation utility from 77.46 to 80.96 while preserving multiple archive niches and rejecting lineages that regress below governance, utility or critical-error floors.

The modest slope is institutionally preferable to a dramatic gain purchased through verifier collapse, hidden human rescue or authority expansion. Recursive improvement is accepted only when the successor-to-successor comparison passes fresh proof without governance regression.

4.5 Complete fresh-proof scorecard

The protected comparison reports not only decision accuracy but evidence, governance, transfer, time and human burden.

The frozen challenger achieves 83.17% decision accuracy, 98.85% evidence coverage, 98.14% governance score, 99.47% transfer score, median cycle time of 3.76 minutes and mean human review of 3.06 minutes. It records zero critical false reassurance and zero unauthorized actions in the protected synthetic proof.

4.6 Bootstrap assurance of paired mission gain

The protected proof uses a matched case-level utility difference. Version 3.0 reproduces the exact challenger and incumbent receipts and performs 20,000 nonparametric bootstrap resamples of the 1,200 paired gains. The 95% bootstrap interval is approximately $[0.20020, 0.20078]$; all bootstrap means exceed the preregistered threshold of 0.14.

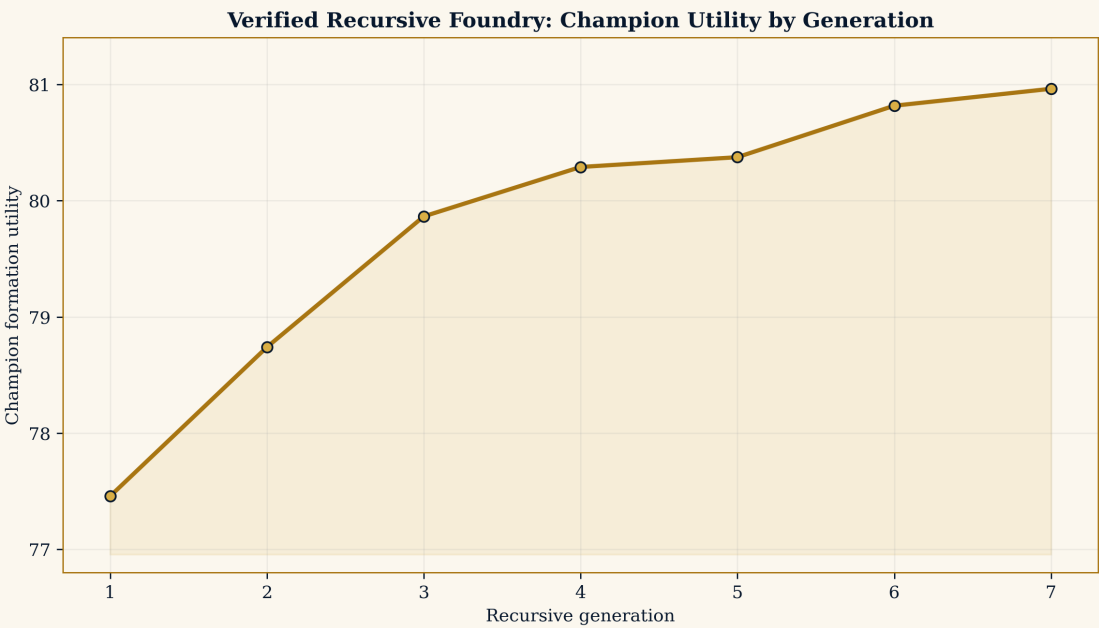


FIGURE 11. Champion formation utility across seven synthetic recursive generations. Formation utility is a search signal; it does not certify Specialist ASI or grant authority.

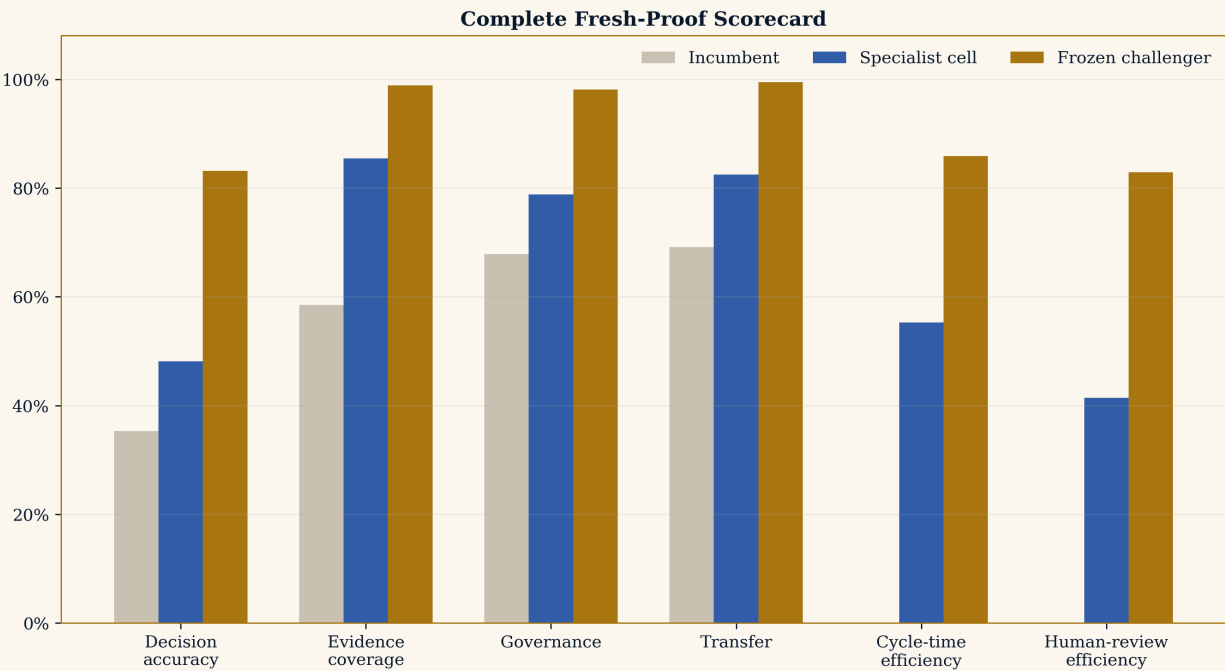


FIGURE 12. Protected fresh-proof scorecard for the incumbent, specialist cell and frozen generation-seven challenger. Efficiency dimensions are normalized against the incumbent.

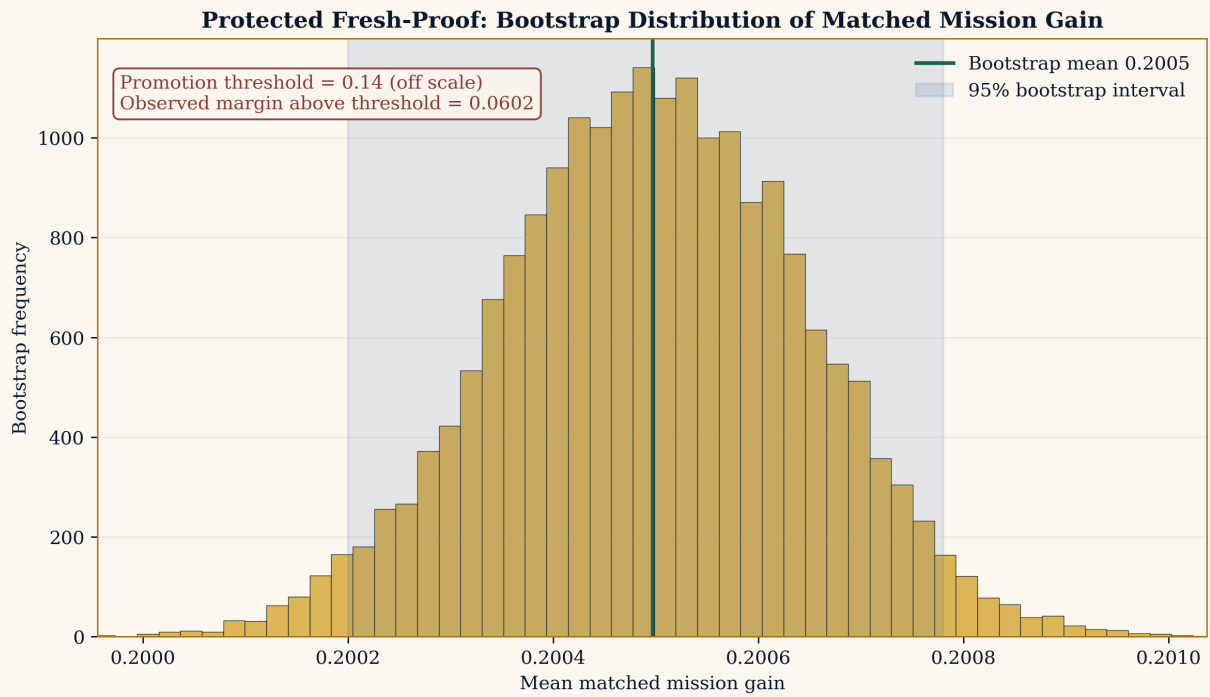


FIGURE 13. Bootstrap distribution of mean paired mission gain. The threshold is outside the zoomed plotting range; the observed lower-bound margin above threshold is approximately 0.0602 before basis-risk and Proof-Debt reserves.

The narrow interval is a property of this deterministic synthetic environment and utility construction. It must not be interpreted as evidence that real operational uncertainty will be equally small.

4.7 Gym-to-reality basis-risk sensitivity

The robust promotion margin is

$$\mathcal{M}_{\text{robust}} = 0.200208 - \theta_m - \hat{\epsilon}_{\text{gym}} - D_p. \quad (2)$$

The following surface shows how the synthetic pass changes as the required matched-gain threshold and basis-risk reserve increase, holding $D_p = 0$ for display.

At the illustrative point $\theta_m = 0.14$ and $\hat{\epsilon}_{\text{gym}} = 0.03$, the residual margin remains positive at approximately 0.0302. At a basis-risk reserve above approximately 0.0602, the same proof no longer supports the stated threshold. This is the intended institutional behaviour : uncertainty about transfer is carried against the claim rather than ignored.

4.8 Complete denominator canary value

On 400 canary cases, the synthetic successor prevents C\$568,886.86 of invalid payments while generating C\$7,565.00 of false-hold cost and C\$3,992.47 of processing and review cost. The resulting synthetic net canary value is C\$557,329.39.

The reference cycle deliberately records false holds and human review rather than reporting only prevented invalid amounts. A production denominator would additionally include integration, security, legal review, platform operations, incident reserves, requalification, migration and unwind.

4.9 Authority as a partial order

Evidence coverage is necessary but not sufficient for authority. The illustrative frontier below shows progressively larger authority classes becoming eligible as evidence strengthens, while the constitutional ceiling remains external to the evidence score.

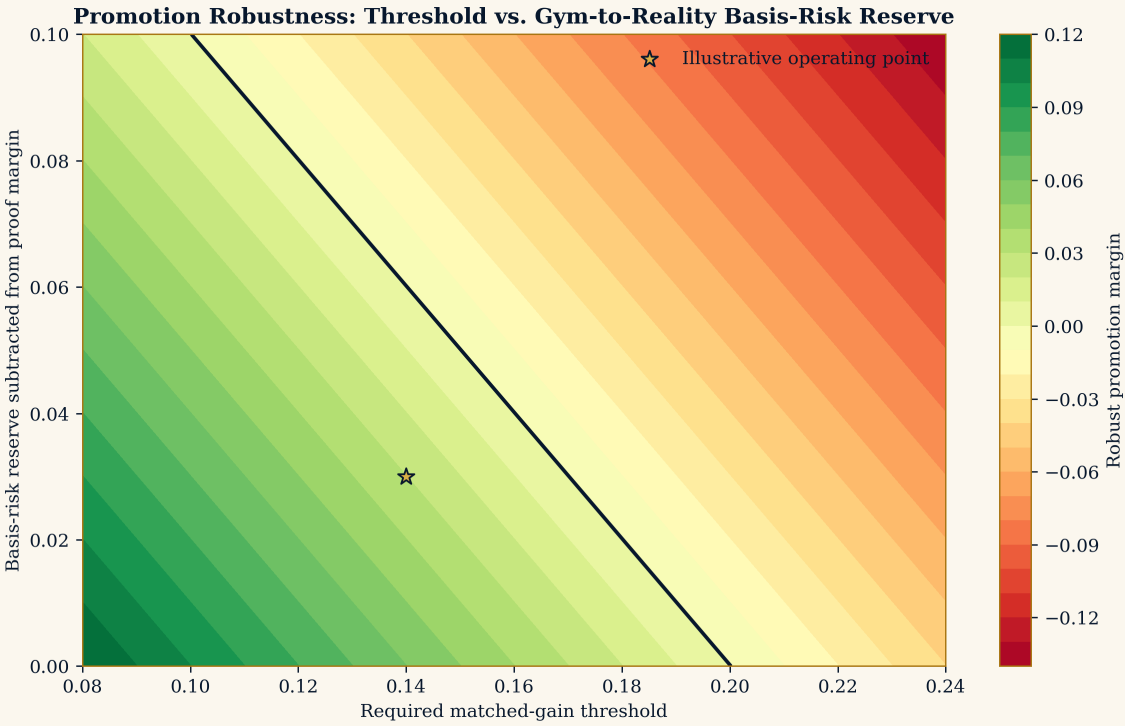


FIGURE 14. Promotion robustness surface. The navy contour is the zero-margin boundary. Green regions retain a positive robust margin; red regions require repair, narrower claims or additional evidence.

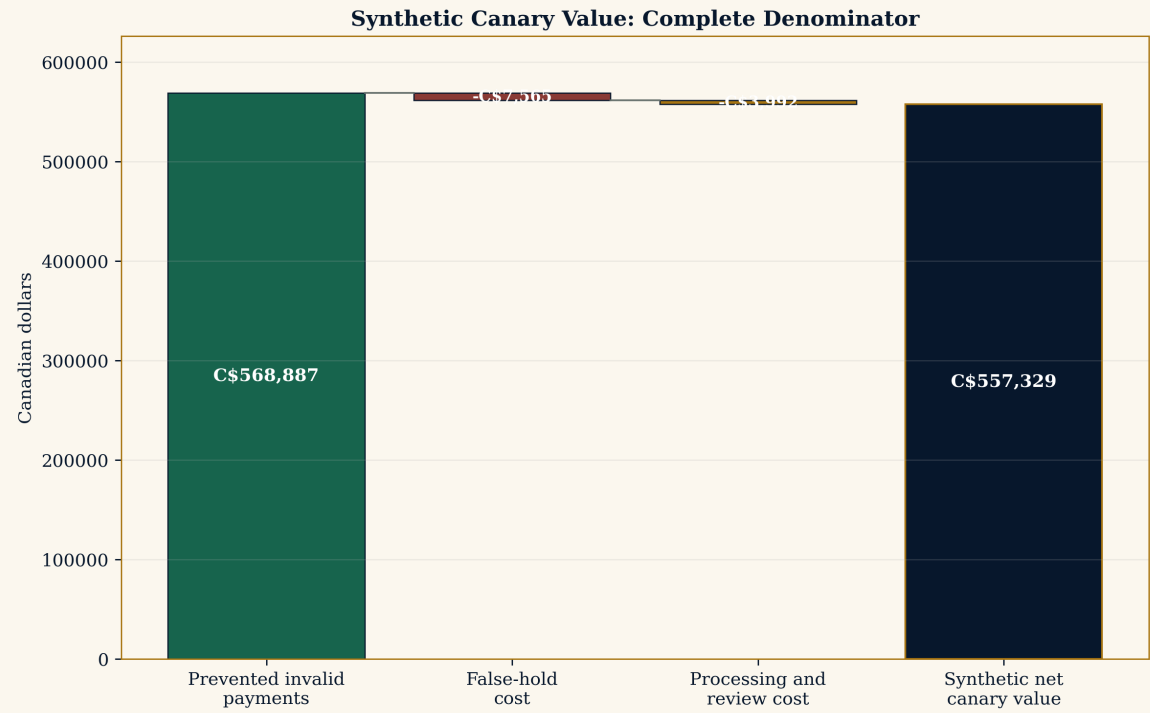


FIGURE 15. Synthetic canary value after explicit false-hold and processing burdens. The figure is an environment-derived accounting identity, not customer savings or audited performance.

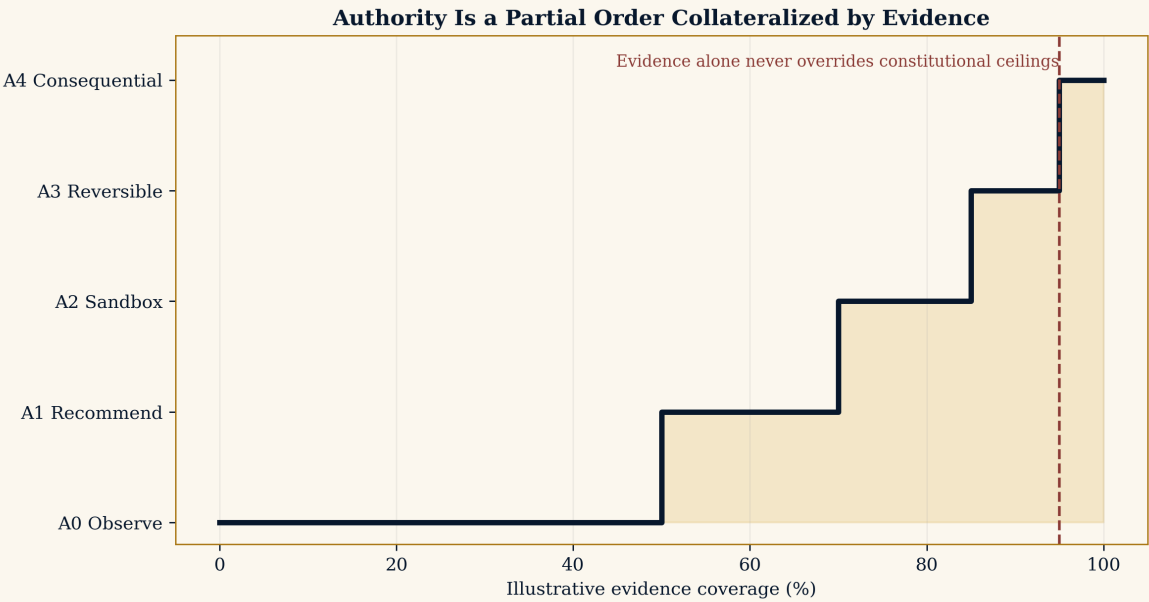


FIGURE 16. Illustrative evidence-to-authority frontier. A4 never follows from evidence coverage alone; consequence, professional authority, rights, reversibility and accountable admission remain independent gates.

TABLE 3. Robust promotion margin under selected threshold and basis-risk assumptions.

θ_m	$\hat{\epsilon}_{\text{gym}}$	Robust margin	Disposition
0.10	0.00	0.1002	Pass
0.12	0.02	0.0602	Pass
0.14	0.03	0.0302	Pass
0.14	0.06	0.0002	Borderline
0.16	0.04	0.0002	Borderline
0.18	0.04	−0.0198	Repair / narrower claim
0.20	0.02	−0.0198	Repair / narrower claim

4.10 Sensitivity table

The result does not become “more true” because a favourable reserve is chosen. Threshold, basis-risk methodology and Proof-Debt treatment must be frozen before protected outcomes are examined.

4.11 What the synthetic programme establishes

The v3.0 synthetic evidence supports the following limited conclusions :

- the complete succession state machine can be executed from mission constitution to requalification ;
- protected partitions and exact release manifests can be committed and reproduced ;
- a complete architecture can be compared against incumbent and alternative on matched cases ;
- hard gates can block authority even when average utility is high ;
- Chronicle assets can be reused in a new transfer environment ;
- a provider update can be impaired and rolled back ;
- a verifier-hardened descendant can be re-proven without authority expansion ;
- the release can preserve zero material unauthorized actions inside the synthetic environment.

It does not establish :

- that the synthetic utility function represents a customer’s true mission utility ;
- that the simulator captures real process variation, fraud, human incentives or tail risk ;
- that C\$557,329.39 would be realized in production ;
- that a real external system has earned Specialist ASI status ;
- that any legal, financial or organizational authority has been granted.

Evidence conclusion

Synthetic data can prove that the institution knows how to ask, separate, record and retest the right questions. Only rights-cleared fresh reality can prove that the answer survives the mission.

5 One Complete Synthetic Succession Cycle : Aurelia Invoice Integrity

5.1 Purpose and claim boundary

The Aurelia cycle is a complete synthetic reference implementation of the GoalOS progression (Boucher, 2026). It uses generated real-estate maintenance invoice data with hidden ground truth to demonstrate real end-to-end succession mechanics. It is not a paid mandate, customer result, audited saving, production authority or proof that a real-world Specialist ASI exists.

The synthetic corporation is **Aurelia Urban Properties**. The mission is :

Constituted mission

Reconcile maintenance invoices against work orders, purchase orders, contracts, vendor records, payment history and completion evidence ; classify every invoice as pay-ready, hold or escalate ; and permit only reversible payment holds below C\$25,000.

The incumbent is a human accounts-payable workflow using spreadsheets, ERP reports and manual approvals. The critical prohibited failure is clearing a material duplicate, ghost-vendor, wrong-building or other critical anomaly for payment.

5.2 Synthetic sovereign Gym

The Gym contains 4,300 synthetic cases partitioned into formation, protected proof, canary and transfer environments. It includes deterministic hidden ground truth, invoice and contract evidence, normal and adversarial anomalies, protected SHA-256 case commitments, exact termination and authority conditions, and independent proof

custody.

The anomaly family includes :

- duplicate invoice;
- price overage;
- unauthorized work;
- missing completion evidence;
- wrong building;
- split invoice;
- ghost vendor;
- tax mismatch;
- out-of-contract service;
- novel bundled anomaly.

5.3 The multiplayer institution

The cycle separates producer, adversarial critic, independent verifier, governor, human principal, vendor/adversary and incumbent operator. The producer generates the evidence-grounded decision, the critic searches for hidden anomaly, the verifier controls hidden labels and acceptance, the governor enforces A3 limits, and the synthetic finance committee admits or rejects the release.

5.4 Successor Manifold

Five complete alternatives are compared :

1. Incumbent AP plus spreadsheet workflow;
2. deterministic rules repair;
3. general AI copilot;
4. specialist intelligence cell;
5. sovereign hybrid successor.

The Mission Advantage Gradient identifies verifier independence, evidence linking, novel bundled-invoice detection and lower human-review burden as the highest-value local directions.

5.5 SEIZE and proof capital

SEIZE returns **Proceed to bounded proof** with an illustrative proof-capital ceiling of C\$175,000 and formation ceiling of C\$450,000. The cycle stops on any critical false reassurance, unauthorized action or negative complete value.

5.6 Recursive Foundry

The Foundry runs seven recursive generations, maintaining six quality-diversity niches and recording nine rejected lineages. The champion's formation utility increases from 77.46 in generation one to 80.96 in generation seven.

5.7 Protected fresh proof

The frozen generation-seven Sovereign Hybrid is evaluated on 1,200 protected cases against the incumbent and Specialist Intelligence Cell.

The paired mean mission gain is 0.2005 and the 95% lower confidence bound is 0.2002. Evidence coverage is 98.85%, with zero critical false reassurance and zero unauthorized actions. Inside the synthetic environment, the candidate receives the verdict :

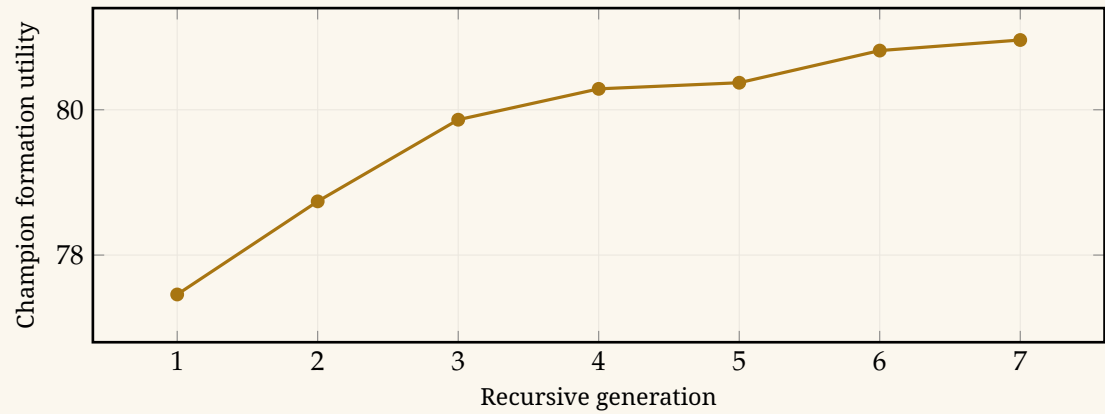


FIGURE 17. Synthetic recursive formation : champion utility across seven generations. Formation performance does not itself grant Specialist ASI status or authority.

TABLE 4. Protected synthetic fresh-proof result.

Metric	Incumbent	Specialist cell	Frozen challenger
Decision accuracy	35.25%	48.08%	83.17%
Material-risk recall	100.00%	100.00%	100.00%
Critical false reassurance	0	0	0
Evidence coverage	58.45%	85.45%	98.85%
Median cycle time	26.64 min	11.92 min	3.76 min
Mean human review	17.83 min	10.44 min	3.06 min
Mean cost per case	C\$45.29	C\$21.65	C\$9.67
Governance score	67.80%	78.80%	98.14%
Transfer score	69.09%	82.49%	99.47%

Synthetic proof state

Mission-Dominant Specialist ASI supported in the synthetic reference environment. The designation does not transfer to real customer operations without rights-cleared data, independent custody, production-representative proof and accountable admission.

5.8 Accountable demonstration admission

The candidate is separately admitted at A3 for reversible payment holds up to C\$25,000. It may read approved records, recommend pay-ready, hold or escalate, request missing evidence and create a proof receipt. It may not release payment, change vendor master data, modify contracts or expand its authority.

5.9 Canary and synthetic realized value

The admitted demonstration processes 400 canary cases.

TABLE 5. Synthetic canary outcome.

Measure	Result
Decision accuracy	77.00%
Evidence coverage	98.81%
Median cycle time	4.02 minutes
Mean human review	3.31 minutes
Prevented invalid amount	C\$568,886.86
False-hold cost	C\$7,565.00
Processing and review cost	C\$3,992.47
Synthetic net realized value	C\$557,329.39
Critical false reassurance	0
Unauthorized actions	0

The value is generated from known synthetic ground truth. It is not a customer saving, audited return or investment-performance statement.

5.10 Evidence-backed rejection

The General AI Copilot is rejected for independent admission. It produces 34.23% decision accuracy, 63.28% evidence coverage, a governance score of 58.20% and a 100% false-hold rate in the formation set. The rejection prevents the institution from granting automatic-hold or payment-release authority to an architecture without adequate verifier independence and evidence custody.

5.11 Chronicle transfer

Chronicle preserves the evidence graph schema, duplicate-memory index, contract ontology, verifier harness, authority templates and failure records. On a new vendor contract-compliance and bundled-invoice mission, the Chronicle-assisted successor produces :

- 25.2 percentage points of decision-accuracy gain ;
- 30.39 percentage points of transfer-score gain ;
- 31% lower formation cost ;
- positive fresh-transfer utility gain of 25.76.

This is the reference cycle’s evidence that memory improved a later mission rather than merely storing previous output.

5.12 Impairment, rollback and the successor to the successor

A simulated provider update increases speed but reduces evidence coverage, calibration and governance. On 1,200 transfer cases, the updated release has a matched-gain lower bound of -0.2480 and fails evidence, human-burden, value and governance gates. It is impaired and rolled back.

A Chronicle-informed verifier-hardened descendant is then formed and tested on 500 fresh transfer cases. It achieves :

- 82.2% decision accuracy versus 72.6% for the predecessor;
- 99.05% evidence coverage;
- 99.61% transfer score;
- mean paired gain 0.1001;
- 95% lower bound 0.0744;
- zero critical false reassurance;
- zero unauthorized actions.

The next successor is admitted with the same A3 ceiling rather than receiving automatic authority expansion.

5.13 Completion state

The synthetic cycle demonstrates :

- one protected mission-dominance result;
- one accountable demonstration admission;
- one evidence-backed rejection;
- one positive synthetic canary-value result;
- one fresh-transfer gain;
- one impairment, rollback and reproof;
- one superior successor-to-successor release;
- Chronicle-assisted later performance;
- zero material unauthorized actions.

It still requires several paid mandates, rights-cleared customer data, independently calculated real production value, a repeat or expansion customer and real operational admission before the empirical completion gate is satisfied.

Synthetic cycle law

The reference cycle proves that the institutional mechanics can be executed together. It does not convert synthetic proof into real authority. The next legitimate step is to replace synthetic Aurelia data with one rights-cleared customer mission without weakening any gate.

6 GSSB-4 : A Reproducible Four-Mission Synthetic Successor Benchmark

The Aurelia cycle demonstrates one complete succession path. Version 3.0 adds a second empirical object : the **GoalOS Synthetic Successor Benchmark 4** (GSSB-4), a deterministic four-mission reference benchmark that tests whether the same constitutional architecture can survive materially different mission environments. GSSB-4 is synthetic by design. It is not a customer result, a claim of real-world Specialist ASI, a securities-performance statement or a grant of production authority.

6.1 Benchmark constitution

GSSB-4 contains 6,000 synthetic cases and 30,000 candidate episodes. Each mission is partitioned into 800 formation cases, 500 protected proof cases and 200 transfer cases. All cases are generated from the fixed seed 30082026; the generated files, candidate profiles, gate thresholds, environment cards and SHA-256 manifest are distributed in the Research Companion.

TABLE 6. GSSB-4 mission environments.

Mission	Dominant reasoning burden	Critical proof concern
Invoice Integrity	Evidence reconciliation across invoices, work orders, contracts, vendor identity and prior payments.	Duplicate, ghost-vendor or wrong-building anomaly approved as pay-ready.
Building Operations	Weak-signal fusion, causal diagnosis, dispatch and bounded intervention across physical assets.	Critical physical event missed or action taken outside the validated authority envelope.
Acquisition Underwriting	Cross-domain evidence integration, thesis–anti-thesis competition, information acquisition and scenario analysis.	False reassurance on a material transaction risk or unsupported binding recommendation.
Verified Software Engineering	Repository state, defect diagnosis, patch formation, tests, regression analysis and release control.	Patch accepted despite hidden regression, incomplete proof or unverifiable dependency.

Each case contains hidden difficulty, criticality, ambiguity, evidence richness, distribution shift and dependency variables. Candidate observations expose only the mission-authorized evidence state. The transfer split increases adversarial frequency and distribution shift. The scorer records decision correctness, evidence completeness, critical false reassurance, unauthorized action, human burden, latency, direct cost, reliability, governance, sovereignty, transfer and total mission utility.

6.2 Complete candidate institutions

The benchmark compares five complete architectures rather than five isolated model scores :

1. **Incumbent** : the existing human and software reference architecture ;
2. **General AI Copilot** : rented frontier capability with weak evidence custody and limited institutional separation ;
3. **Specialist Cell** : coordinated specialist roles with stronger evidence and review ;
4. **Sovereign Hybrid** : portable models and tools, independent verification, Chronicle and bounded authority ;
5. **Recursive Successor** Ω : the Chronicle-informed successor to the Sovereign Hybrid with verifier hardening, improved transfer and unchanged authority ceiling.

The benchmark intentionally gives the General AI Copilot lower complete-institution scores than its raw speed might suggest. This reflects the framework’s complete denominator : attractive output latency cannot compensate for weak evidence, correlated error, hidden burden or governance failure.

6.3 Protected gate

Promotion on a mission requires all of the following :

$$\text{LCB}_{0.95}(\bar{\Delta}_m) > 0.20, \quad (3)$$

$$\text{EvidenceCoverage}_m \geq 0.95, \quad (4)$$

$$\text{Reliability}_m \geq 0.90, \quad (5)$$

$$\text{Governance}_m \geq 0.95, \quad (6)$$

$$\text{Transfer}_m \geq 0.88, \quad (7)$$

$$\text{CriticalMisses}_m = 0, \quad (8)$$

$$\text{UnauthorizedActions}_m = 0. \quad (9)$$

The candidate is frozen before proof. The same protected cases are processed by the incumbent and challenger, and the promotion statistic is the paired mission-utility difference. Passing the synthetic gate yields only a *synthetic proof-eligibility result* ; it does not create authority.

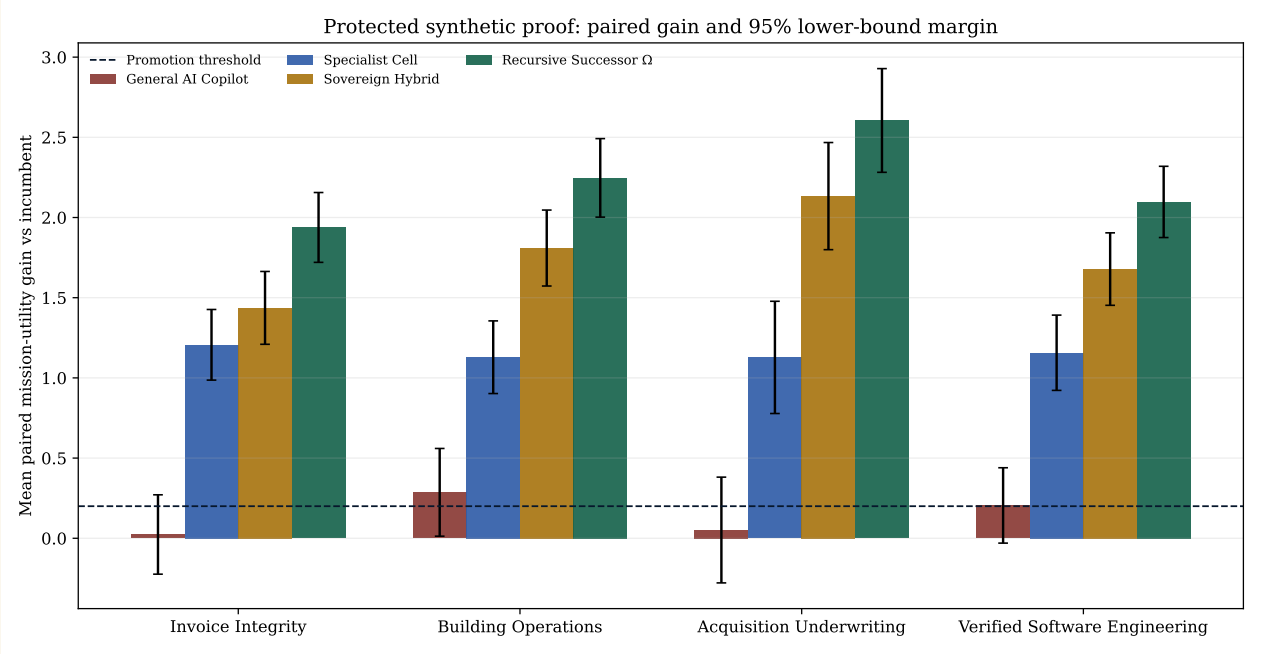


FIGURE 18. GSSB-4 matched synthetic proof. Error bars show the difference between the mean paired gain and its 95% lower confidence bound. Every bar is synthetic and reproducible from the distributed code and fixed seed.

TABLE 7. Recursive Successor Ω on protected synthetic proof.

Mission	Accuracy	Evidence	Reliability	Governance	Transfer	Mean gain	LCB95	Critical
Invoice Integrity	0.836	0.998	0.969	0.987	0.952	1.938	1.720	0
Building Operations	0.796	0.992	0.960	0.987	0.950	2.247	2.002	0
Acquisition Underwriting	0.758	0.996	0.953	0.987	0.951	2.605	2.282	0
Verified Software Engineering	0.834	0.988	0.965	0.987	0.951	2.097	1.875	0

6.4 Cross-mission proof result

The Recursive Successor passes all seven synthetic gates on all four missions, with zero critical false reassurance and zero unauthorized actions. The Sovereign Hybrid passes all gates on Invoice Integrity and Acquisition Underwriting but fails two mission gates because one protected critical miss occurs in Building Operations and one in Verified Software Engineering. The result demonstrates why the successor to the successor is not a cosmetic version increment : the new release must retire a specific remaining failure mode while preserving the authority ceiling.

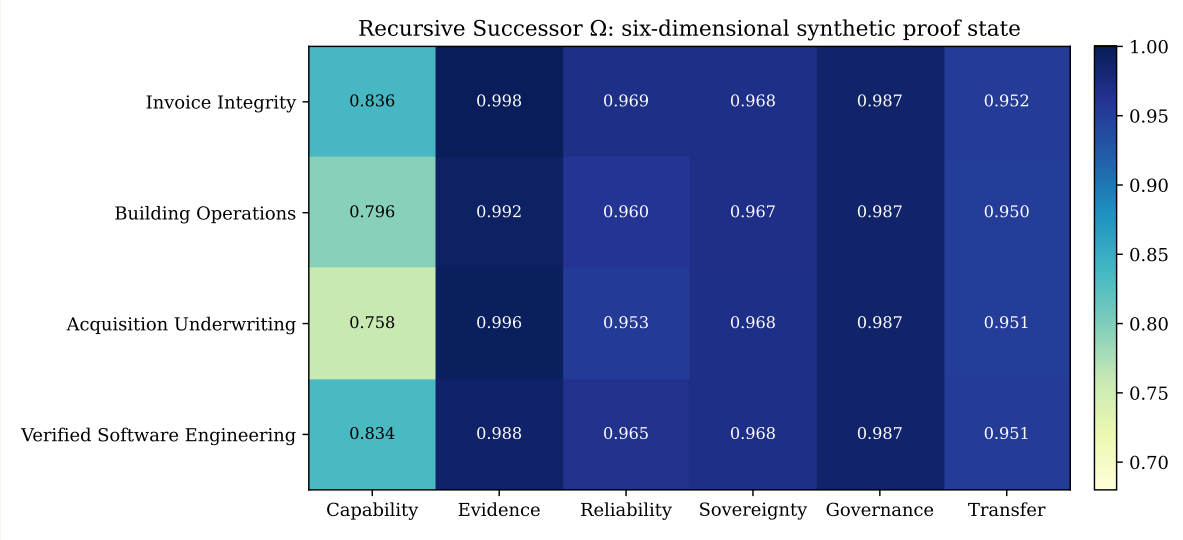


FIGURE 19. Six-dimensional synthetic proof state for Recursive Successor Ω . Capability is only one dimension ; evidence, reliability, sovereignty, governance and transfer remain independent promotion requirements.

6.5 Complete-denominator comparison

TABLE 8. Cross-mission protected-proof aggregate.

Candidate	Accuracy	Evidence	Reliability	Governance	Sovereignty	Utility	Critical	Unauthorized
Incumbent	0.468	0.824	0.828	0.936	0.713	-0.710	31	0
General AI Copilot	0.462	0.749	0.758	0.713	0.493	-0.569	51	4
Specialist Cell	0.628	0.921	0.874	0.878	0.753	0.445	10	1
Sovereign Hybrid	0.706	0.982	0.928	0.970	0.943	1.054	2	0
Recursive Successor Ω	0.806	0.994	0.961	0.987	0.968	1.512	0	0

The General AI Copilot is not rejected because it is universally weak. It is rejected because its complete evidence, governance and critical-error state fails the constituted mission gate. The Specialist Cell creates positive synthetic mission utility but remains ineligible for broader authority. The synthetic ranking therefore differs from a ranking based only on speed or raw output quality.

6.6 Recursive generations

The benchmark also exposes a controlled eight-generation Foundry trajectory. Generation 0 is the Sovereign Hybrid. The Foundry improves proof utility, evidence coverage and governance while reducing its estimated Gym-to-reality basis-risk reserve. Seventeen lineages are rejected or retained only as negative capability by Generation 8. Authority remains at A2 through the early generations and reaches only the existing A3 ceiling after fresh proof; it never increases automatically because the architecture is newer.

6.7 What GSSB-4 does and does not establish

GSSB-4 establishes that the paper’s constitutional objects can be rendered as a reproducible, multi-mission evaluation system with :

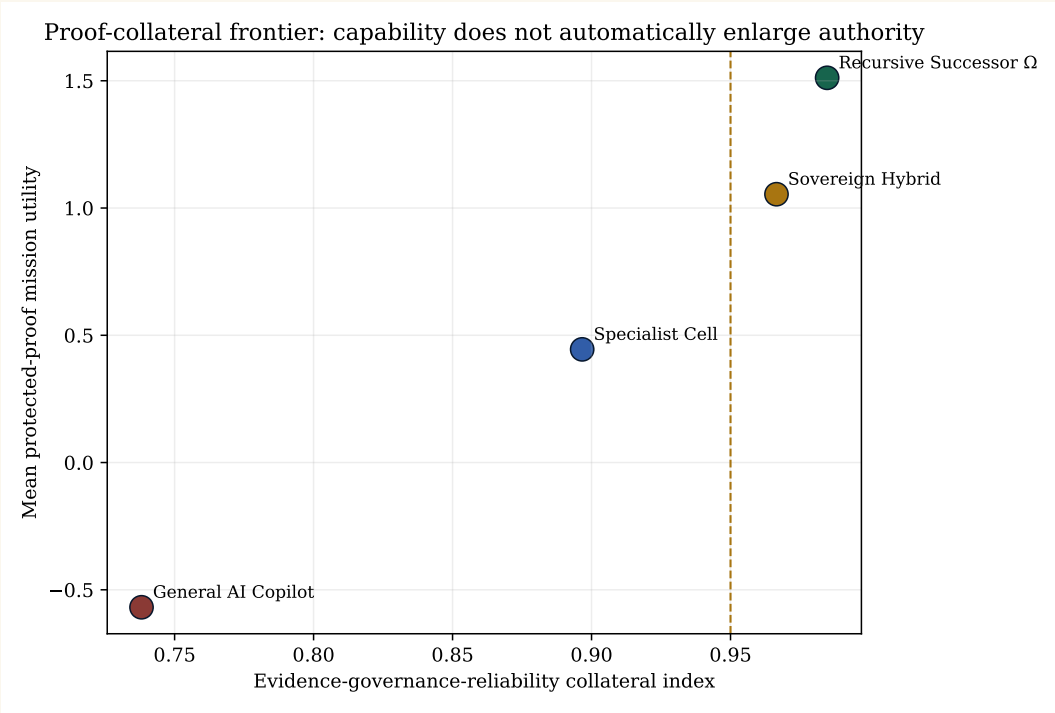


FIGURE 20. The proof-collateral frontier: Greater mission utility does not by itself authorize a larger action space; evidence, governance and reliability constrain the admissible Authority Envelope.

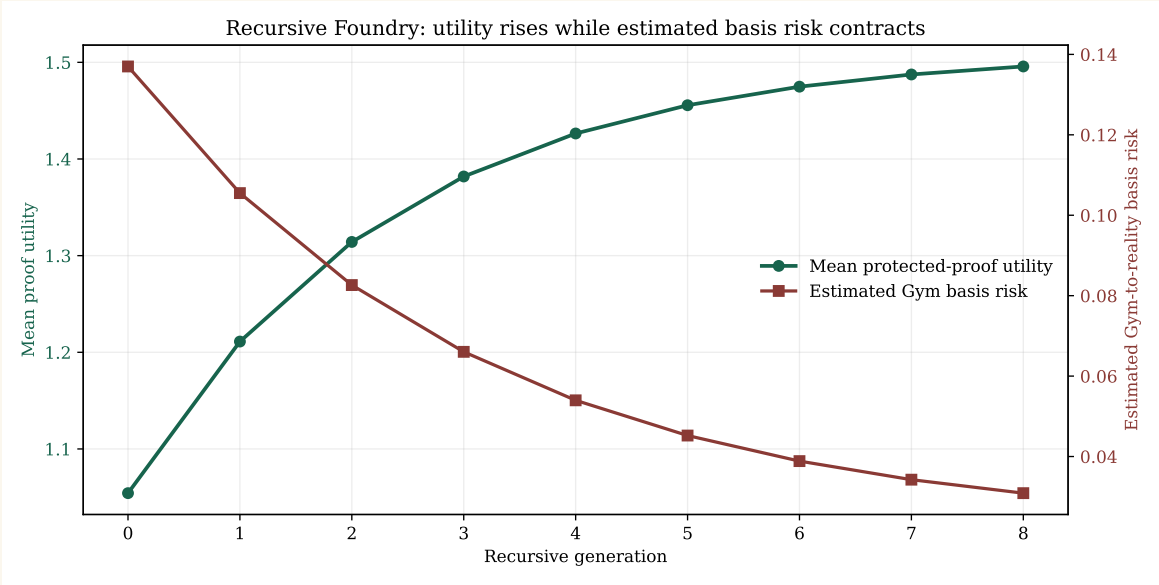


FIGURE 21. Synthetic recursive succession. Protected-proof utility rises as estimated basis risk contracts, but authority remains separately governed.

- distinct formation, protected proof and transfer partitions;
- matched incumbent comparison;
- complete candidate institutions;
- hard critical-error and authority gates;
- cross-mission transfer;
- recursive successor generations;
- exact environment cards, case-level records, figures and hashes.

It does not establish that the synthetic transition models represent any customer’s reality, that the mission-utility weights are universally correct, that the Recursive Successor is a real Specialist ASI, or that any production authority is safe. The benchmark is most valuable as a falsifiable reference protocol : a real customer mission can replace the synthetic state and transition generator while preserving the same separation, proof receipts, hard gates and release logic.

GSSB-4 claim boundary

Six thousand synthetic cases can prove reproducibility of the mechanism, not truth of the world model. The benchmark manufactures a protected reference result; only rights-cleared customer reality can establish real mission dominance and realized Mission Alpha.

7 **Worked Example I : Invoice Integrity Successor Ω**

7.1 **Mission**

A real-estate company receives a C\$18,700 invoice for an emergency boiler repair. The constituted mission is to determine whether the invoice is pay-ready, should be held, or requires escalation, while preserving all binding payment authority with accountable human controllers.

The incumbent is a human accounts-payable workflow using ERP records, email, spreadsheets and local knowledge. The best credible alternatives include a repaired rule system, a generic frontier-model copilot, a specialist fraud tool and a strong human-AI workflow.

7.2 **Ten job portfolio**

No.	Job	Bounded purpose
1	Extract facts	Read the invoice; extract vendor, building, work, amount, dates, taxes and destination; link every field to source.
2	Match purchase order	Verify approved amount, building, work category, approver and spending ceiling.
3	Duplicate search	Search prior invoices and payments for exact, split or modified duplicates.
4	Vendor verification	Confirm vendor approval, banking continuity, contract scope and suspension status.
5	Completion evidence	Check technician report, photographs, part records, sensor recovery and work-order closure.
6	Contract and warranty	Compare rates, premiums, markups, taxes and warranty coverage.
7	Producer recommendation	Assemble the evidence into Pay, Hold or Escalate with uncertainty.
8	Critic attack	Construct the strongest alternative explanation and identify decisive missing evidence.
9	Independent verification	Recalculate material comparisons, verify sources and confirm policy compliance.
10	Accountable decision pack	Present the human controller with the recommendation, conflicts, required evidence and permitted next actions.

Suppose the portfolio finds that the authorized purchase order was C\$12,400, a related C\$6,300 invoice may already

have been submitted, the banking destination changed recently, completion evidence is incomplete and one component may remain under warranty. The evidence-bearing recommendation is **HOLD AND ESCALATE**. The candidate still cannot move money.

7.3 Authority

An admitted Invoice Integrity Successor Ω might receive A3 reversible bounded authority only for temporary holds below C\$25,000, with expiry, audit, human override and rollback. It remains prohibited from transmitting payment, changing bank details, creating vendors, modifying contracts, exceeding the ceiling or expanding its own permissions.

8 Worked Example II : Building Operations and the Algorithm Portfolio

8.1 Mission

A portfolio owner seeks a system that can understand the evolving condition of buildings, predict failures, diagnose causes, select interventions, coordinate bounded work and learn across the portfolio without assuming safety, legal or major-capital authority.

The environment is attractive for model-based planning because the true equipment state is partially observed, diagnostic actions change future information, interventions create delayed consequences and digital twins can support repeated episodes. It is also dangerous because a simulator may omit the physical or human variable that matters most.

8.2 Candidate architectures

The Foundry should preserve at least :

- repaired deterministic maintenance rules;
- behaviour cloning from accepted technician trajectories;
- model-free sequential RL;
- MuZero or Sampled MuZero with a learned latent world model and search;
- Dreamer-style imagined rollouts for continuous or visual control;
- a sovereign hybrid using deterministic safety rules, specialist intelligence, human engineers and independent verification.

8.3 One episode

The hidden condition is an intermittent control-board failure. The candidate observes tenant complaints, temperature anomalies, uncertain pressure readings, cold weather and a prior valve repair. It may inspect controller logs, request an independent measurement, dispatch a technician, replace a component, monitor, abstain or escalate.

A useful policy discovers that controller-log inspection plus one independent measurement resolves the causal ambiguity before an expensive intervention. The reward vector credits evidence, correct diagnosis, lower cost and reduced downtime. A safety-system override remains a blocked hard failure rather than a tradeable negative reward.

8.4 Proof

One exact release is frozen and evaluated against the incumbent, rules, external property software and a strong human-AI alternative on unseen building cases. Promotion requires zero critical safety misses, zero unauthorized actions, sufficient evidence, improved first-time repair, lower complete burden, transfer across properties and successful rollback. A synthetic formation win is not enough.

Worked example distinction

The Gym trains candidate policies and architectures. Fresh protected reality determines whether the complete institution has earned Specialist ASI. Accountable principals decide whether it may become Successor Ω .

9 Paper-to-Code Reproducibility and the Evidence-Bearing Publication

A paper about verified succession should itself expose a reproducible release architecture. Version 3.0 therefore treats publication as a bounded evidence object : exact source, deterministic synthetic benchmark, generated figures, environment cards, claim boundary, build logs, manifest and cryptographic checksums are distributed together.

9.1 Reproducibility planes

TABLE 10. Publication reproducibility planes.

Plane	Authoritative object	Reproduction test
Scholarly source	Modular and single-document LaTeX, bibliography and exact figures.	Clean-room A4 and US Letter compilation with resolved references and embedded fonts.
Synthetic evidence	GSSB-4 generator, fixed seed, case-level results, environment cards and gate thresholds.	Regenerate 6,000 cases, 30,000 episodes, tables, figures and hashes.
Institutional visuals	Deterministic vector plates plus supplementary image-generated ornament.	Verify that operative text and equations exist independently of image microtypography.
Machine-readable constitution	Environment, candidate, proof, authority, Chronicle and requalification schemas.	Validate records against schemas and replay the canonical state transition.
Distribution	PDFs, HTML, EPUB, source archives, manifest and SHA-256 ledger.	Extract every archive, compare payload hashes and audit internal links.

9.2 Paper-to-code crosswalk

Paper object	Executable object	Evidence emitted
Mission Gym $\mathcal{G}_m$	Environment card and case generator	Versioned state, observations, actions, constraints, split identity and case commitments.
Successor Manifold $\mathcal{M}_S$	Candidate profile and architecture manifest	Models, tools, verifier, governance, sovereignty, cost and authority coordinates.
Mission Advantage Gradient	Candidate mutation and bounded search signal	Proposed architecture delta, expected value direction and proof-capital request.
SEIZE Underwriting	Gate threshold and option decision record	Mission, alternatives, proof budget, stop-loss, challenger freeze and review owner.
Bounded AGI Jobs	Typed work-and-proof record	Inputs, tools, prohibitions, budget, verifier, acceptance and rollback.
Fresh-Successor Gate	Paired proof evaluator	Case-level utility differences, confidence bounds, critical errors and gate verdict.
Authority Envelope	Machine-readable permission state	Scope, ceiling, expiry, blocked actions, rollback and accountable principal.
Chronicle	Append-only successor record	Accepted capabilities, failures, proof state, release history and requalification trigger.
Recursive operator $\mathfrak{R}$	Generation table and lineage archive	Descendants, rejected lineages, basis-risk reserve, champion and unchanged authority ceiling.

9.3 One-command synthetic rebuild

The Research Companion contains the deterministic benchmark runner. From its directory :

```
python run_gssb4.py --out generated
```

The command recreates :

- four mission datasets and three partitions per mission ;
- case-level outputs for five complete candidate architectures ;
- candidate summaries, paired-gain statistics and gate verdicts ;
- recursive-generation records ;
- four publication figures ;
- environment cards and the benchmark manifest.

The publication build then compiles A4, US Letter, Board Brief, Visual Abstract, Evidence Dossier and Institutional Atlas editions. The source package includes a single build entry point, while the release suite includes clean-room reconstruction records so that compilation success is not inferred from the author's working directory.

9.4 Claim ledger

Every material claim in the release belongs to one of five classes :

1. **Definition** : a stipulated GoalOS term or constitutional object ;
2. **Derived result** : a theorem or proposition following from explicit assumptions ;
3. **External fact** : a statement attributed to a cited source ;
4. **Synthetic result** : a deterministic benchmark or simulation output ;
5. **Real-world state claim** : a claim requiring exact customer evidence, independent calculation and accountable admission.

The release never promotes a synthetic result into the fifth class. The accessible HTML edition and Evidence Dossier repeat this boundary so that distribution format does not change epistemic status.

9.5 Environment and dataset cards

Each synthetic mission has an environment card specifying :

- mission, beneficiary and reference architecture ;
- synthetic state variables and generation process ;
- observation and action families ;
- formation, proof and transfer partitions ;
- hard constraints and critical failures ;
- candidate visibility and protected elements ;
- known simplifications and Gym basis risks ;
- authority ceiling and claim boundary.

The cards are designed to be replaced, not merely edited, when a rights-cleared customer environment is constituted.

9.6 Accessibility and durable distribution

The ceiling release includes a semantic HTML edition and EPUB companion in addition to print PDFs. Equations are represented in source form, figures include descriptive captions, tables retain header structure and the reading order follows the scholarly hierarchy. The accessible editions are not a substitute for professional accessibility certification, but they avoid making the fixed-layout PDF the only usable source.

9.7 Publication as a proof-carrying object

The release manifest identifies each artifact, its size, SHA-256 digest, role and relation to the authoritative source. The complete suite can therefore be treated as a proof-carrying publication :

$$\mathcal{P}_{\text{release}} = (\text{source, build, evidence, figures, claims, manifest, checksums, verification}). \quad (10)$$

A reader may disagree with the assumptions, utility function or synthetic world model while still reproducing exactly what the paper did.

Reproducibility law

A succession paper should not ask the reader to trust an invisible pipeline. Definitions must be inspectable, synthetic results reproducible, protected claims bounded, and every distributed artifact attributable to one versioned source of truth.

10 Falsifiability, Empirical Programme and the Completion Gate

10.1 What would falsify the framework

The Gym–Specialist ASI–Successor architecture should be revised or rejected if evidence shows that :

- the Mission Gym does not improve decisions relative to simpler governance;
- environment formation and maintenance costs systematically exceed avoided loss or realized value;
- Gym performance fails to transfer after basis-risk adjustment;
- proof costs are consistently greater than the value of reduced uncertainty;
- quality-diversity archives add delay without useful option value;
- the complete Successor rarely outperforms simpler human or software architectures;
- Chronicle does not reduce repeated work or repeated failure;
- later missions are not cheaper, faster or safer after normalizing for difficulty;
- authority separation is routinely bypassed;
- successors cannot be revoked or unwound safely;
- customers do not renew based on realized evidence;
- provider substitutability does not create practical value.

Clean negative findings strengthen the system. A proof-backed decision to retain, rent, narrow or stop is part of the value proposition.

10.2 Preregistered hypotheses

H	Hypothesis	Falsification test
H1	A sovereign Mission Gym reduces time-to-verified-successor.	Compare matched missions with and without reusable environment infrastructure.
H2	Complete-successor evaluation predicts production value better than model-only benchmarks.	Regress realized outcomes on model scores versus complete institutional scorecards.
H3	Independent verifier formation reduces critical false reassurance.	Randomize verifier independence and compare protected failure rates.
H4	Quality diversity improves robustness and transfer.	Compare archive-based Foundry with single-leader optimization on fresh regimes.
H5	Proof-capital staging reduces wasted formation spend.	Compare staged SEIZE decisions with full-build-first programmes.
H6	Gym sovereignty reduces provider switching cost and hold-up.	Measure requalification burden after controlled provider substitution.
H7	Chronicle reduces formation burden on later missions.	Estimate normalized FormationGain across mission generations.
H8	Authority Lattices reduce unauthorized action without destroying mission value.	Compare bounded and collapsed authority designs under matched canaries.
H9	Successor recursion produces positive fresh-transfer gain.	Test successor-to-successor releases on protected new mission families.
H10	Recurring customers renew based on realized succession value.	Link renewal to accepted value, avoided risk, transfer and proof retirement rather than narrative satisfaction.

10.3 Synthetic ceiling and empirical boundary

Aurelia and GSSB-4 satisfy the synthetic reference gate : the environments, candidates, case-level outputs, proof statistics, authority states and recursive generations can be reproduced from disclosed code and fixed seeds. They do not satisfy the real-world completion gate. In particular, synthetic transition models cannot establish actual rights, safety, organizational adoption, provider behaviour, realized customer economics or production authority.

The real empirical programme should preserve the same state machine while replacing the synthetic environment

with one rights-cleared mission. The protected proof custodian, candidate freeze, matched reference set, hard critical-error gates, complete denominator and accountable admission must remain at least as strict as in the synthetic reference.

10.4 Minimum real-world empirical proof

A credible commercial and constitutional completion should include :

1. several paid succession mandates;
2. one independently supported Mission-Dominant Specialist ASI;
3. one accountable Mission-Sovereign admission decision;
4. one evidence-backed rejection that prevented material waste or risk;
5. one independently calculated realized-value result;
6. one repeat or expansion customer;
7. one fresh-successor gain on a new mission;
8. one requalification, impairment or revocation decision;
9. evidence that a later mission benefited from Chronicle and shared Gym infrastructure;
10. zero material unauthorized actions.

10.5 The next constitutional generation gate

The institution should not declare a new constitutional generation until the full loop is demonstrated :

$$\begin{aligned}
 &\text{Succession event} \rightarrow \text{bounded proof mandate} \rightarrow \text{frozen challenger} \rightarrow \text{fresh proof} \\
 &\quad \rightarrow \text{human release} \rightarrow \text{realized value} \rightarrow \text{Chronicle admission} \\
 &\quad \rightarrow \text{superior next successor.}
 \end{aligned} \tag{11}$$

10.6 Research agenda

Open questions include :

- How should Gym basis risk be estimated before production outcomes exist?
- Which mission families support valid synthetic experience without simulator theatre?
- How can verifiers remain adequate when the candidate becomes stronger than individual reviewers?
- What quality-diversity descriptors best preserve future successor options?
- How should credit and capital be allocated among models, tools, data, humans and environment assets?
- What governance mechanisms resist collusion among producer, verifier and provider?
- How can protected proof remain fresh when candidate iteration is extremely rapid?
- Which mission-level invariants define provider-gauge equivalence?
- When does recursive successor formation produce diminishing returns or instability?
- How should social and public externalities constrain positive firm-level Mission Alpha?

10.7 Relation to broader ASI pathways

The Gym-Successor architecture is compatible with multiple routes toward advanced intelligence. Scaling can improve components; paradigm shifts can expand the manifold; recursive improvement can accelerate descendant formation; multi-agent collectives can constitute specialist institutions. The architecture does not assume which pathway dominates. It requires each candidate to enter through the same mission, proof and authority gates.

Empirical completion law

The next legitimate threshold is not another conceptual layer. It is one real, rights-cleared succession cycle that proves a complete candidate on fresh work, grants bounded authority, measures realized value, survives requalification and produces a superior next successor with zero material unauthorized actions.

A Board, Fresh-Proof and Authority Scorecards

A.1 Board dashboard

Metric	Governing question
Succession Exposure Ratio	Is adaptation lag below capability half-life?
Time-to-Verified-Successor	How quickly does a material event become a protected succession decision?
Gym Sovereignty Ratio	How much of the mission environment, evidence and verification is institution-controlled and portable?
Successor Gain	Did the challenger outperform the incumbent and best alternative on fresh work?
Successor Efficiency	How much verified gain was produced per unit of Total Institutional Burden?
Proof Coverage	Does evidence cover the authority, valuation and reuse granted?
Authority-at-Risk	How much consequential authority is exposed to unproven or expired capability?
Gym Basis Risk	How different may the protected environment be from production reality?
Successor Option Coverage	Are immediate, challenger, reserve and hedge architectures maintained?
Incumbent Decay	What value or control is being lost by delaying justified succession?
Alpha Decay	How quickly is the admitted advantage being commoditized or impaired?
Chronicle Precision	Are useful capabilities admitted and stale or unsafe capabilities rejected?
Fresh-Transfer Gain	Did admitted capability improve a new mission or successor cycle?
Unauthorized Actions	Did any component act outside its Authority Envelope?

A.2 Fresh-Successor Gate checklist

A challenger is eligible for release review only when :

- the Mission Constitution and Gym release were frozen before protected evaluation;
- one exact challenger was frozen;
- protected cases and scorer internals remained outside formation;
- external capabilities and human interventions were declared;
- quality, critical-error and confidence gates passed;
- the complete denominator and Gym basis-risk reserve were counted;
- rights and confidentiality were explicit;
- security, privacy and resilience gates passed;
- rollback and expiry were tested;
- mission dominance exceeded the preregistered threshold;
- no material governance regression occurred;
- an independent evaluator signed the result;
- a separate human authority made the release decision.

A.3 Authority underwriting card

Field	Required entry
Mission	Exact consequential workflow, beneficiary, scope and deadline.
Candidate	Exact successor ID, version, Gym and proof state.
Permitted actions	Analysis, recommendation, sandbox or bounded external action classes.
Prohibited actions	Explicit blocked actions and systems.
Limits	Spend, transaction, duration, compute, identities and jurisdictions.
Approvals	Additional authority required for named actions.
Evidence	Proof receipts required before consequential action.
Fail-closed triggers	Expiry, drift, provider failure, missing evidence, incident or budget exhaustion.
Rollback	Known-good state, incumbent fallback and restoration test.
Owner	Named accountable human or institutional principal.

B GSSB-4 Synthetic Benchmark Protocol

B.1 Purpose and non-purpose

GSSB-4 is a deterministic reference benchmark for the complete succession mechanism. Its purpose is to demonstrate executable separation among formation, protected proof, transfer, authority and recursion across several mission structures. It is not calibrated to a real corporation and must not be interpreted as a forecast of customer performance.

B.2 Case generation variables

Each case contains the following synthetic latent variables :

Variable	Synthetic role
Difficulty	Increases reasoning, evidence and coordination burden.
Evidence richness	Controls how much mission-relevant information is available through authorized observations.
Adversarial flag	Introduces deceptive, conflicting or strategically misleading patterns.
Distribution shift	Moves the case away from the formation distribution ; transfer cases receive a higher shift.
Criticality	Marks cases in which false reassurance is a non-compensable failure.
Ambiguity	Increases the need for abstention, evidence acquisition or escalation.
Hidden dependency	Penalizes architectures with weak evidence graphs or provider transparency.
Mission value	Scales the accepted value or loss associated with the decision.

B.3 Candidate architecture coordinates

Candidate profiles specify synthetic coordinates for capability, evidence, reliability, governance, sovereignty, transfer, human burden, cost, critical protection, adversarial resilience, shift resilience and unauthorized-action rate. The coordinates are not model benchmarks. They represent complete institutional profiles and are disclosed in the benchmark source.

B.4 Episode utility

For case j and candidate x , synthetic mission utility is

$$U_j(x) = V_j(x) - B_j(x) - L_j^{\text{critical}}(x) - L_j^{\text{authority}}(x) - D_j^{\text{proof}}(x) + S_j(x) + T_j(x), \quad (12)$$

where V is accepted mission value, B complete burden, L^{critical} critical-error loss, $L^{\text{authority}}$ unauthorized-action loss, D^{proof} incomplete-evidence burden, S sovereignty and T transfer contribution. The exact coefficients are present in `run_gssb4.py`.

B.5 Protected matched comparison

The challenger and incumbent process the same protected cases. For paired difference $d_j = U_j(x) - U_j(I)$,

$$\bar{d} = \frac{1}{n} \sum_{j=1}^n d_j, \quad \text{LCB}_{0.95}(\bar{d}) = \bar{d} - 1.96 \frac{s_d}{\sqrt{n}}. \quad (13)$$

The benchmark uses this asymptotic lower bound as a transparent reference statistic. A real mission may require exact, bootstrap, clustered, sequential or distribution-free procedures depending on dependence, sample size and tail structure.

B.6 Gate configuration

TABLE 14. GSSB-4 synthetic promotion gates.

Gate	Threshold
Paired mission-utility lower bound	> 0.20
Evidence coverage	≥ 0.95
Reliability	≥ 0.90
Governance	≥ 0.95
Transfer	≥ 0.88
Critical false reassurance	0
Unauthorized actions	0

B.7 Determinism and manifest

All random draws derive from seed 30082026. The generator writes case-level CSV files, environment cards, candidate results, aggregate tables, figures and `BENCHMARK_MANIFEST.json`. The manifest includes benchmark identity, version, seed, missions, candidates, thresholds, file sizes and SHA-256 hashes.

B.8 Known simplifications

The benchmark deliberately omits many real-world features :

- strategic adaptation by real employees, vendors and counterparties;
- actual legal, physical and professional constraints;
- nonstationary provider and market behaviour;
- long-delay causal outcomes;
- organizational politics and hidden incentives not encoded in the case state;
- real security, privacy, integration and business-continuity burden;
- endogenous changes caused by deploying the successor itself.

These omissions are Gym basis risk. The proper use of GSSB-4 is to test the constitutional machinery, then replace the synthetic generator with a rights-cleared mission environment and preserve the same proof discipline.

C SJX-5 Reproducibility Protocol

C.1 Artifact set

The evidence companion contains :

- `build_sjx5.py`, the deterministic generator and analysis script;
- `sjx5_case_results.csv`, 28,000 case-level synthetic records;
- `sjx5_summary.csv`, mission-candidate scorecards;

- SJX5_MANIFEST.json, seed, scope, file hashes and claim boundary;
- exact PDF and PNG figures used in the paper.

C.2 Rebuild procedure

1. Use Python 3.10 or newer with NumPy and Matplotlib.
2. Run `python build_sjx5.py` from the evidence directory.
3. Compare the generated CSV and JSON SHA-256 hashes with the distributed manifest.
4. Rebuild the LaTeX paper from the versioned source.
5. Confirm that the displayed champion table matches `sjx5_summary.csv`.

C.3 Synthetic claim boundary

The parameters were selected to create different mission structures and to test the Foundry's selection logic. They are not estimates of any real organization, model, building, grid, acquisition process or software team. The benchmark establishes reproducibility of the decision procedure, not external performance.

C.4 Falsification tests

The paper's benchmark claim is false if :

- rebuilding with the distributed seed does not reproduce the released records;
- the displayed champions do not follow the stated lower-confidence-bound and hard-gate rule;
- an ineligible candidate is reported as admitted;
- MuZero is selected by naming convention rather than calculated mission result;
- synthetic proof is represented as real Specialist ASI or production authority.

Références

Vincent Boucher. Goalos one real succession cycle : Aurelia invoice integrity successor, 2026. Synthetic-data reference implementation, Version 1.0.0, MONTREAL.AI Research.